

SAT UZBEKISTAN

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Module-1

1.

$$(x-6)^2 + (y+5)^2 = 16$$

In the xy -plane, the graph of the equation above is a circle. Point P is on the circle and has coordinates $(10, -5)$. If $\overline{PQ}$ is a diameter of the circle, what are the coordinates of point Q ?

☐ (A) $(2, -5)$

☐ (B) $(6, -1)$

☐ (C) $(6, -5)$

☐ (D) $(6, -9)$

2.

A circle in the xy -plane has its center at $(-5, 2)$ and has a radius of 9. An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. What is the value of c ?

3.

In the xy -plane, a circle has center C with coordinates (h, k) . Points A and B lie on the circle. Point A has coordinates $(h+1, k+\sqrt{102})$, and $\angle ACB$ is a right angle. What is the length of $\overline{AB}$?

☐ (A) $\sqrt{206}$

☐ (B) $2\sqrt{102}$

☐ (C) $103\sqrt{2}$

☐ (D) $103\sqrt{3}$

4.

The equation $x^2 + (y-2)^2 = 36$ represents circle A. Circle B is obtained by shifting circle A down 4 units in the xy -plane. Which of the following equations represents circle B?

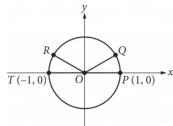
☐ (A) $x^2 + (y+2)^2 = 36$

☐ (B) $x^2 + (y-6)^2 = 36$

☐ (C) $(x-4)^2 + (y-2)^2 = 36$

☐ (D) $(x+4)^2 + (y-2)^2 = 36$

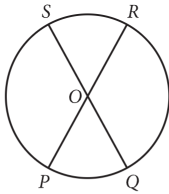
5.



In the xy -plane above, points P , Q , R , and T lie on the circle with center O . The degree measures of angles POQ and ROT are each 30° . What is the radian measure of angle QOR ?

- ☐ (A) $\frac{5}{6}\pi$
- ☐ (B) $\frac{3}{4}\pi$
- ☐ (C) $\frac{2}{3}\pi$
- ☐ (D) $\frac{1}{3}\pi$

6.



Note: Figure not drawn to scale.

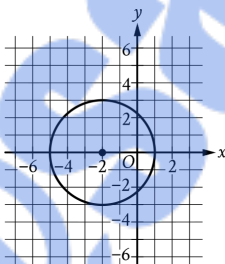
The circle shown has center O , circumference 144π , and diameters $\overline{PR}$ and $\overline{QS}$. The length of arc PS is twice the length of arc PQ . What is the length of arc QR ?

- ☐ (A) 24π
- ☐ (B) 48π
- ☐ (C) 72π
- ☐ (D) 96π

7.

Points A and B lie on a circle with radius 1, and arc $\widehat{AB}$ has length $\frac{\pi}{3}$. What fraction of the circumference of the circle is the length of arc $\widehat{AB}$?

8.



Circle A (shown) is defined by the equation $(x + 2)^2 + y^2 = 9$. Circle B (not shown) is the result of shifting circle A down 6 units and increasing the radius so that the radius of circle B is 2 times the radius of circle A . Which equation defines circle B ?

- ☐ (A) $(x + 2)^2 + (y + 6)^2 = (4)(9)$
- ☐ (B) $2(x + 2)^2 + 2(y + 6)^2 = 9$
- ☐ (C) $(x + 2)^2 + (y - 6)^2 = (4)(9)$
- ☐ (D) $2(x + 2)^2 + 2(y - 6)^2 = 9$

9. A circle in the xy -plane has equation $(x+3)^2 + (y-1)^2 = 25$. Which of the following points does NOT lie in the interior of the circle?

- ☐ A $(-7, 3)$
- ☐ B $(-3, 1)$
- ☐ C $(0, 0)$
- ☐ D $(3, 2)$

10.

A circle in the xy -plane has its center at $(-4, 5)$ and the point $(-8, 8)$ lies on the circle. Which equation represents this circle?

- ☐ A $(x-4)^2 + (y+5)^2 = 5$
- ☐ B $(x+4)^2 + (y-5)^2 = 5$
- ☐ C $(x-4)^2 + (y+5)^2 = 25$
- ☐ D $(x+4)^2 + (y-5)^2 = 25$

11.

$$x^2 + 14x + y^2 = 6y + 109$$

In the xy -plane, the graph of the given equation is a circle. What is the length of the circle's radius?

- ☐ A $\sqrt{109}$
- ☐ B $\sqrt{149}$
- ☐ C $\sqrt{167}$
- ☐ D $\sqrt{341}$

12.

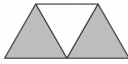
In triangle XYZ , angle Y is a right angle, the measure of angle Z is 33° , and the length of $\overline{YZ}$ is 26 units. If the area, in square units, of triangle XYZ can be represented by the expression $k \tan 33^\circ$, where k is a constant, what is the value of k ?

13.

The perimeter of an isosceles right triangle is $18 + 18\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

- ☐ A 9
- ☐ B $9\sqrt{2}$
- ☐ C 18
- ☐ D $18\sqrt{2}$

14.



A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles. If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

☐ A $2\sqrt{3}$

☐ B $4\sqrt{3}$

☐ C $8\sqrt{3}$

☐ D 16

15.

The perimeter of an equilateral triangle is 624 centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k ?



16.

In triangle ABC , angle B is a right angle. The length of side AB is $10\sqrt{37}$ and the length of side BC is $24\sqrt{37}$. What is the length of side AC ?

☐ A $14\sqrt{37}$

☐ B $26\sqrt{37}$

☐ C $34\sqrt{37}$

☐ D $\sqrt{34 \cdot 37}$

17.

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D , and angles C and F are right angles. If $\cos B = \frac{1}{22}$, what is the value of $\cos E$?

☐ A $\frac{1}{22}$

☐ B $\frac{1}{23}$

☐ C $\frac{21}{22}$

☐ D $\frac{22}{23}$

18.

The function g is defined by $g(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $g(15a)$ and $g(7a)$?



19.

An isosceles right triangle has a hypotenuse of length 58 inches. What is the perimeter, in inches, of this triangle?

☐ A $29\sqrt{2}$

☐ B $58\sqrt{2}$

☐ C $58 + 58\sqrt{2}$

☐ D $58 + 116\sqrt{2}$

20.

A rectangle is inscribed in a circle, such that each vertex of the rectangle lies on the circumference of the circle. The diagonal of the rectangle is twice the length of the shortest side of the rectangle. The area of the rectangle is $1,089\sqrt{3}$ square units. What is the length, in units, of the diameter of the circle?



21.

The perimeter of an equilateral triangle is 852 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w ?



22.

An isosceles right triangle has a perimeter of $94 + 94\sqrt{2}$ inches. What is the length, in inches, of one leg of this triangle?

☐ A 47

☐ B $47\sqrt{2}$

☐ C 94

☐ D $94\sqrt{2}$

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Module-2

1.

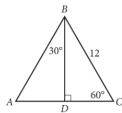
A right triangle has legs with lengths of 24 centimeters and 21 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d ?

2.

Triangle ABC is similar to triangle DEF , where A corresponds to D and C corresponds to F . Angles C and F are right angles. If $\tan(A) = \sqrt{3}$ and $DF = 125$, what is the length of DE ?

- ☐ A $125\frac{\sqrt{3}}{3}$
- ☐ B $125\frac{\sqrt{3}}{2}$
- ☐ C $125\sqrt{3}$
- ☐ D 250

3.



In $\triangle ABC$ above, what is the length of $\overline{AD}$?

- ☐ A 4
- ☐ B 6
- ☐ C $6\sqrt{2}$
- ☐ D $6\sqrt{3}$

4.

The length of a rectangle's diagonal is $3\sqrt{17}$, and the length of the rectangle's shorter side is 3. What is the length of the rectangle's longer side?

5.

Which of the following expressions is equivalent to $(\sin 24^\circ)(\cos 66^\circ) + (\cos 24^\circ)(\sin 66^\circ)$?

- ☐ A $2(\cos 66^\circ)(\sin 24^\circ)$
- ☐ B $2(\cos 66^\circ) + 2(\cos 24^\circ)$
- ☐ C $(\cos 66^\circ)^2 + (\cos 24^\circ)^2$
- ☐ D $(\cos 66^\circ)^2 + (\sin 24^\circ)^2$

6.

A square is inscribed in a circle. The radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. What is the side length, in inches, of the square?

- ☐ A 20
- ☐ B $\frac{20\sqrt{2}}{2}$
- ☐ C $20\sqrt{2}$
- ☐ D 40

7.

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles C and F are right angles. The length of $\overline{AB}$ is 2.9 times the length of $\overline{DE}$. If $\tan A = \frac{21}{20}$, what is the value of $\sin D$?

8.

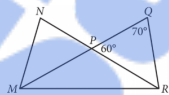
In a right triangle, the tangent of one of the two acute angles is $\frac{\sqrt{3}}{3}$. What is the tangent of the other acute angle?

- ☐ A $-\frac{\sqrt{3}}{3}$
- ☐ B $-\frac{3}{\sqrt{3}}$
- ☐ C $\frac{\sqrt{3}}{3}$
- ☐ D $\frac{3}{\sqrt{3}}$

9.

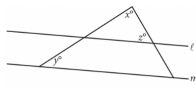
A triangle with angle measures 30° , 60° , and 90° has a perimeter of $18+6\sqrt{3}$. What is the length of the longest side of the triangle?

10.



In the figure above, $\overline{MQ}$ and $\overline{NR}$ intersect at point P , $NP = QP$, and $MP = PR$. What is the measure, in degrees, of $\angle QMR$? (Disregard the degree symbol when gridding your answer.)

11.



Note: Figure not drawn to scale.

In the figure above, lines l and m are parallel, $y = 20$, and $z = 60$. What is the value of x ?

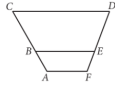
☐ A 120

☐ B 100

☐ C 90

☐ D 80

12.



In the figure above, $\overline{AF}$, $\overline{BE}$, and $\overline{CD}$ are parallel. Points B and E lie on $\overline{AC}$ and $\overline{FD}$, respectively. If $AB = 9$, $BC = 18.5$, and $FE = 8.5$, what is the length of $\overline{ED}$, to the nearest tenth?

☐ A 16.8

☐ B 17.5

☐ C 18.4

☐ D 19.6

13.

In triangle ABC , the measure of angle B is 90° and $\overline{BD}$ is an altitude of the triangle. The length of $\overline{AB}$ is 15 and the length of $\overline{AC}$ is 23 greater than the length of $\overline{AB}$. What is the value of $\frac{BC}{BD}$?

☐ A $\frac{15}{38}$

☐ B $\frac{15}{23}$

☐ C $\frac{23}{15}$

☐ D $\frac{38}{15}$

14.

In triangle DEF , the measure of angle D is 47° and the measure of angle E is 97° . In triangle RST , the measure of angle R is 47° and the measure of angle S is 97° . Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST ?

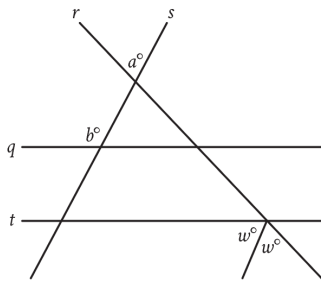
☐ A The measure of angle F

☐ B The measure of angle T

☐ C The measure of angle F and the measure of angle T

☐ D No additional information is needed.

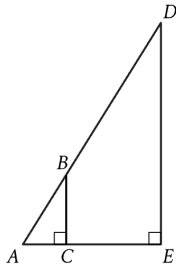
15.



Note: Figure not drawn to scale.

In the figure, parallel lines q and t are intersected by lines r and s . If $a = 43$ and $b = 122$, what is the value of w ?

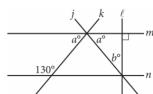
16.



Note: Figure not drawn to scale.

In the figure shown, $AB = \sqrt{34}$ units, $AC = 3$ units, and $CE = 21$ units. What is the area, in square units, of triangle ADE ?

17.

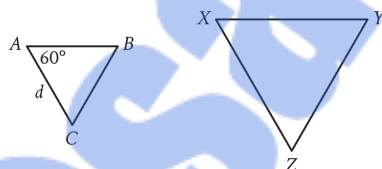


Note: Figure not drawn to scale.

In the figure above, lines m and n are parallel. What is the value of b ?

- ☐ A 40
- ☐ B 50
- ☐ C 65
- ☐ D 80

18.



Note: Figures not drawn to scale.

For the triangles shown, triangle ABC is dilated by a scale factor of 3 to obtain triangle XYZ , where $d = 16$. What is the measure, in degrees, of angle X ?

- ☐ A 20
- ☐ B 57
- ☐ C 60
- ☐ D 63

19.

In triangle ABC , the measure of angle A is 54° , the measure of angle B is 90° , and the measure of angle C is $\left(\frac{k}{2}\right)^\circ$. What is the value of k ?

- ☐ A 36
- ☐ B 45
- ☐ C 72
- ☐ D 108

20.

A rectangle has a length that is 15 times its width. The function $y = (15w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of $15w$ in this context?

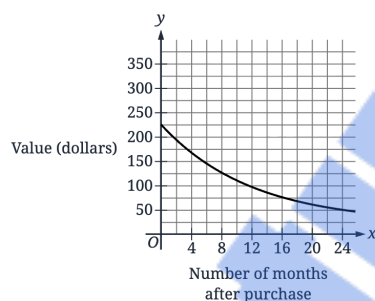
- ☐ A The length of the rectangle, in feet
- ☐ B The area of the rectangle, in square feet
- ☐ C The difference between the length and the width of the rectangle, in feet
- ☐ D The width of the rectangle, in feet

21.

The product of two positive integers is 462. If the first integer is 5 greater than twice the second integer, what is the smaller of the two integers?



22.



The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y-intercept of the graph in this context?

- ☐ A The estimated value of the tablet was \$225 when it was purchased.
- ☐ B The estimated value of the tablet 24 months after it was purchased was \$225.
- ☐ C The estimated value of the tablet had decreased by \$225 in the 24 months after it was purchased.
- ☐ D The estimated value of the tablet decreased by approximately 2.25% each year after it was purchased.